

Welcome To ICCP2024



16th IEEE International
Conference on Computational
Photography (ICCP2024)



July 22-24, 2024

ICCP2024 General and Program Chairs



Sabine Süsstrunk
EPFL



Ashok Veeraraghavan
Rice University



Roarke Horstmeyer
Duke University



Wolfgang Heidrich
KAUST

General Chair

Program Chairs

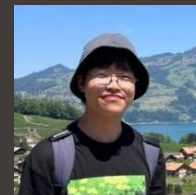
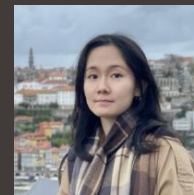
ICCP2024 Organizing team



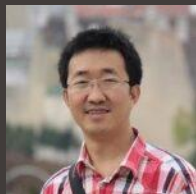
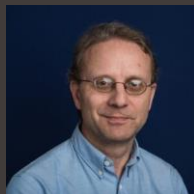
Publication chairs
Vivek Boominathan, Akshat Dave



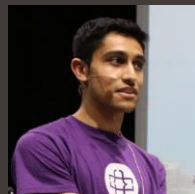
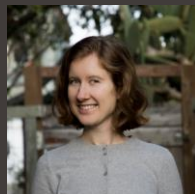
Finance chairs
Suren Jayasuriya, Jinwei Ye



Website chairs
Dongqing Wang, Yufan Ren



Industry chairs
Edoardo Charbon, Feng Yang



Poster & Demo chairs
Kristina Monakhova, Adithya Pediredla, Eric Bezzam



Local arrangements chairs
Raphaël Achddou, Nicoletta Isaac



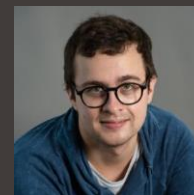
PAMI chair
Boxin Shi



Industry consortium chair
Dengxin Dai



DEI and outreach chairs
Emma Alexander, John Murray-Bruce



Social media chair
Martin N Everaert

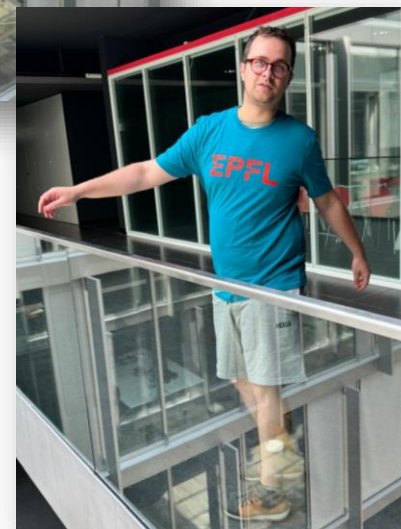
Special thanks!

Raphaël Achddou
EPFL



Yitao Xu
Liying Lu
Zhuoqian Yang
Daichi Zhang
Zhantao Deng
Zicong Jiang

Lovro Nuic
Yufan Ren
Tong Zhang
Bahar Aydemir
Federico Stella
Adrian Jarret
Vasiliki Stergiopoulou



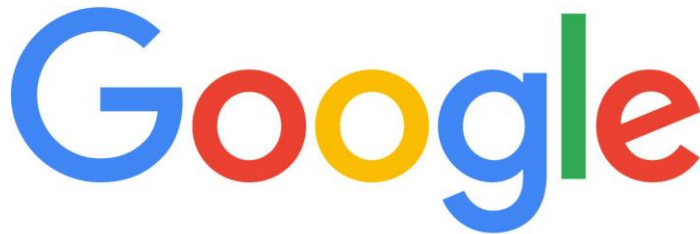
Platinum sponsors:

SONY



HUAWEI

Silver sponsors:



ICCP2024 Sponsors

Bronze sponsors:

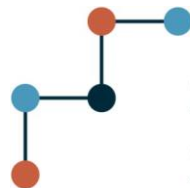
The Omron logo consists of the word "OMRON" in a bold, blue, sans-serif typeface.The Kitware logo features a stylized icon on the left made of several parallel diagonal lines in green and blue, followed by the word "kitware" in a blue, lowercase, sans-serif font.The Ubicept logo includes a circular icon on the left composed of small dots in green, blue, and purple, followed by the word "ubicept" in a black, lowercase, sans-serif font.

Make sure to visit demo booths from our sponsors in SG 0 during the poster and demo sessions!

The Snap Inc. logo is displayed in a bold, black, serif font against a solid yellow rectangular background.The Qualcomm logo is written in a blue, sans-serif font, with the "Q" being notably larger and more stylized than the other letters.

ICCP2024 Sponsors

Student Travel Support:



**Swiss National
Science Foundation**

Travel grants for

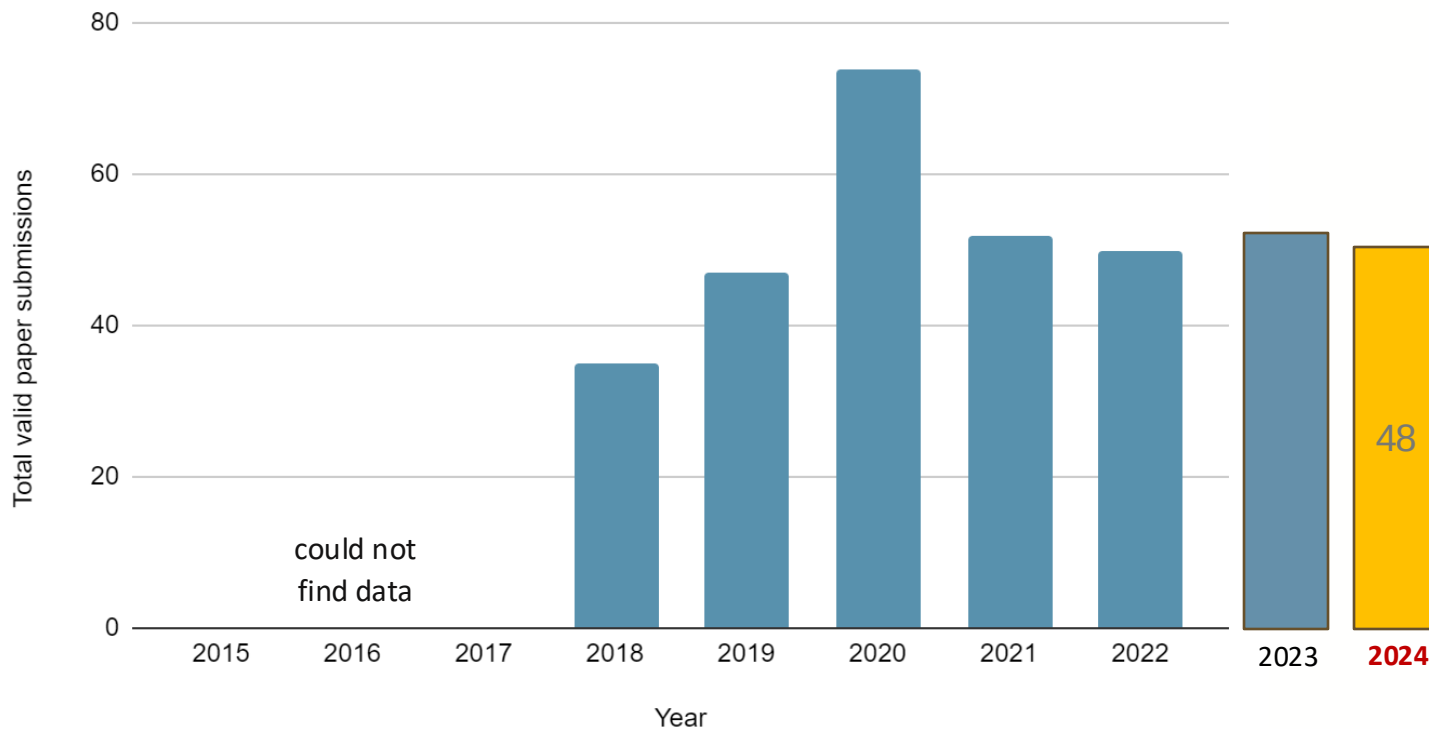
- 10 US-based students
- 4 Asia-based students
- 4 EU-based students

ICCP2024 Sponsors

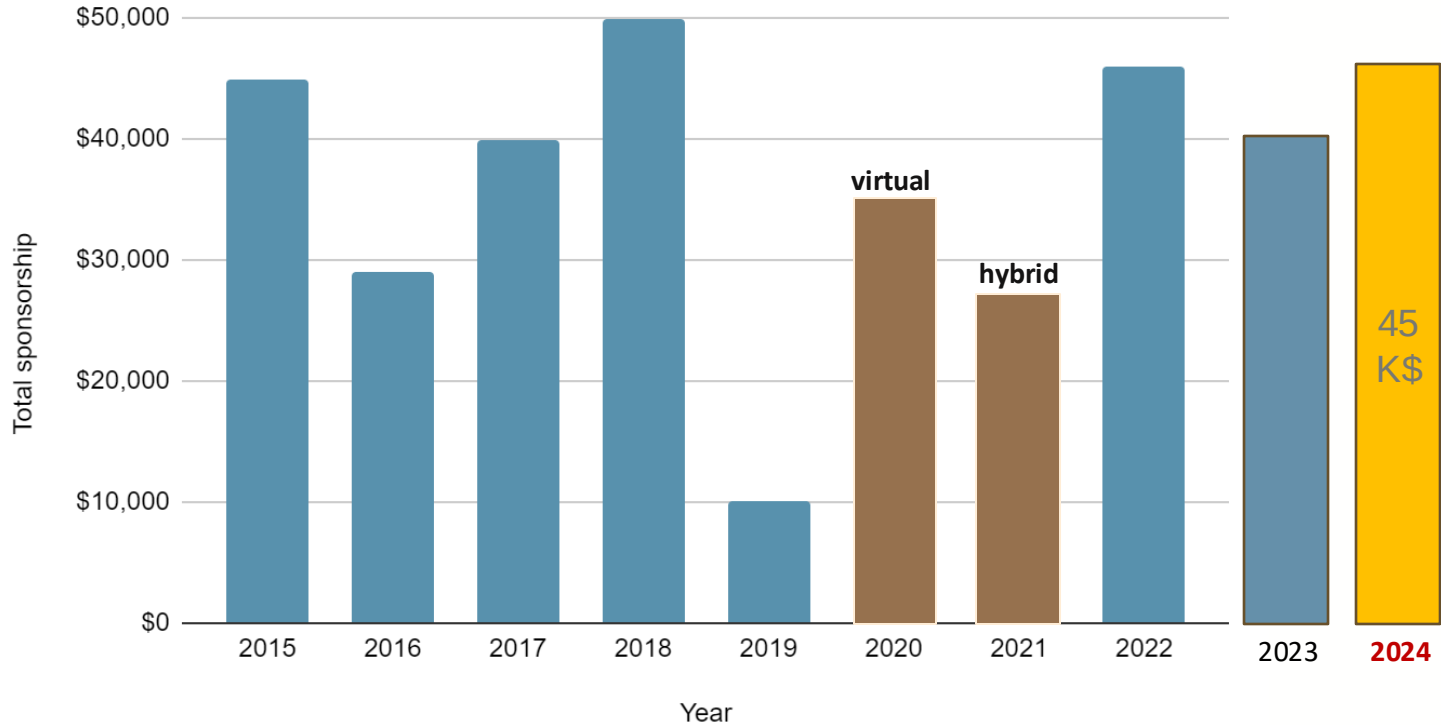


**IEEE
COMPUTER
SOCIETY**

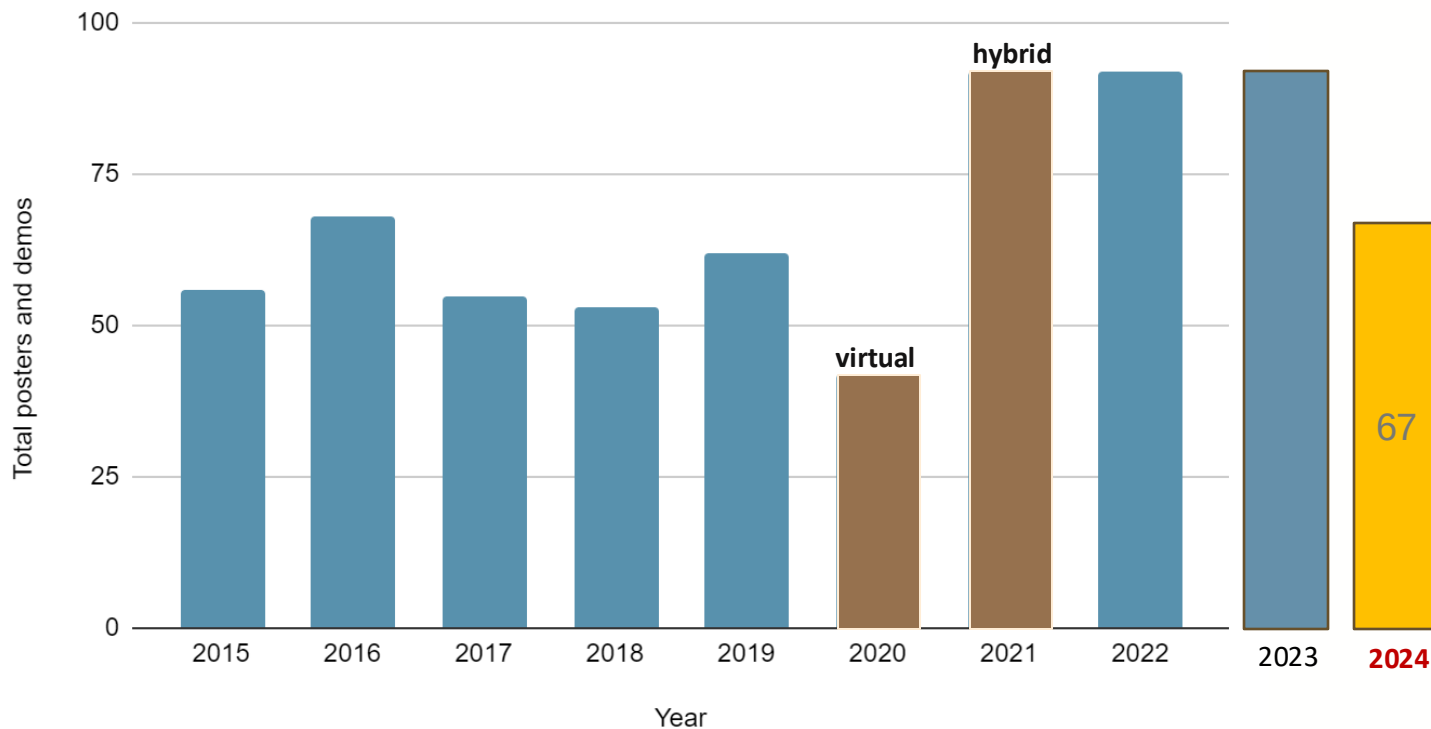
ICCP Paper Submissions Over the Years

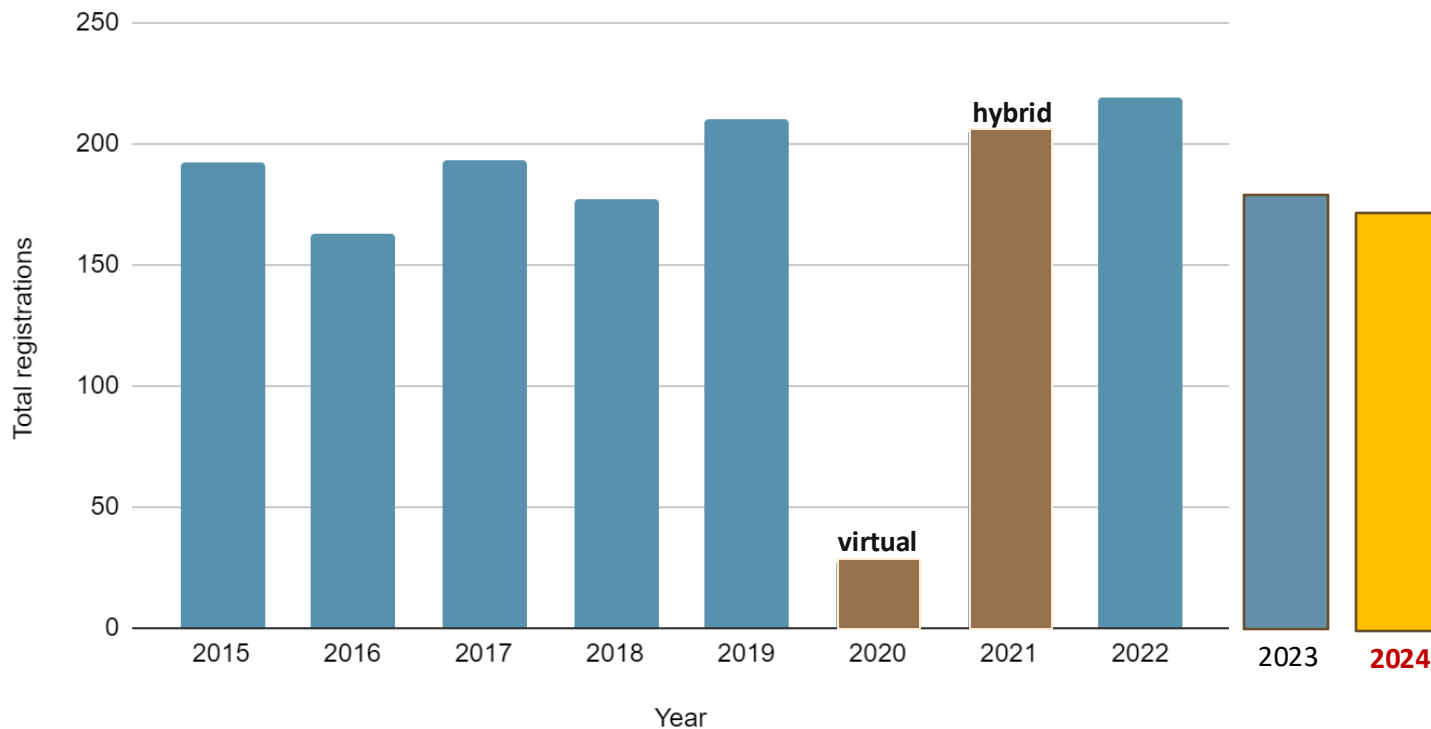


ICCP Sponsorship Over the Years



ICCP Posters/ Demos Over the Years





Lausanne area

Lausanne
city center

Vidy beach

Olympic museum

Ouchy neighborhood



EPFL Metro
Station



SG building



EPFL/Colladon
bus stop

South

North



Migros
+ Denner



SwissTech
Convention Center



SwissTech
Hotel



EPFL Metro
Station



Rolex Learning Center



EPFL
Pavilions



Starling
Hotel



EPFL/Colladon
bus stop



EPFL
Innovation
Park



SG building



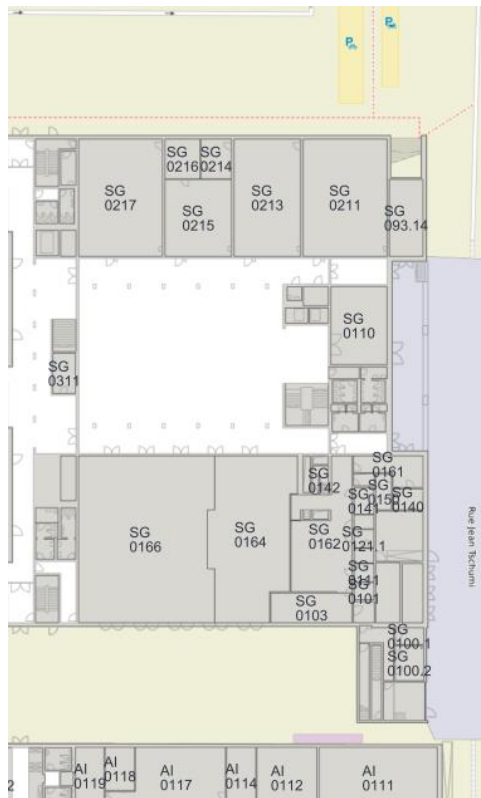
Migros
+ DennerSwissTech
Convention Center

Rolex Learning Center

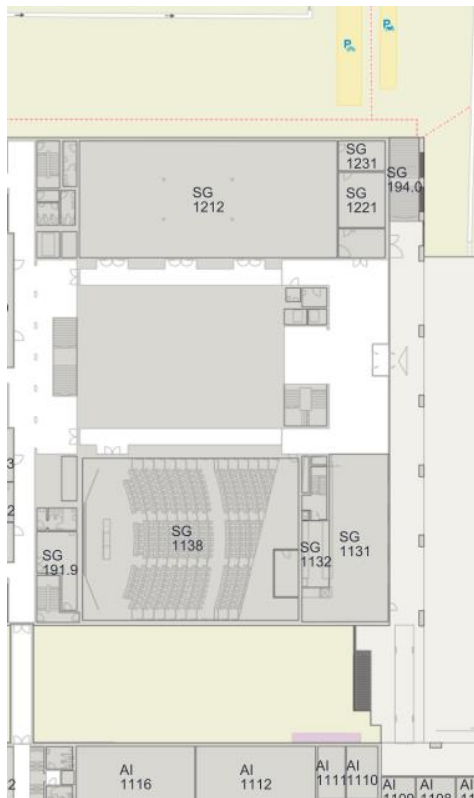
EPFL
PavilionsEPFL
Innovation
ParkSwissTech
HotelEPFL Metro
Station

SG building

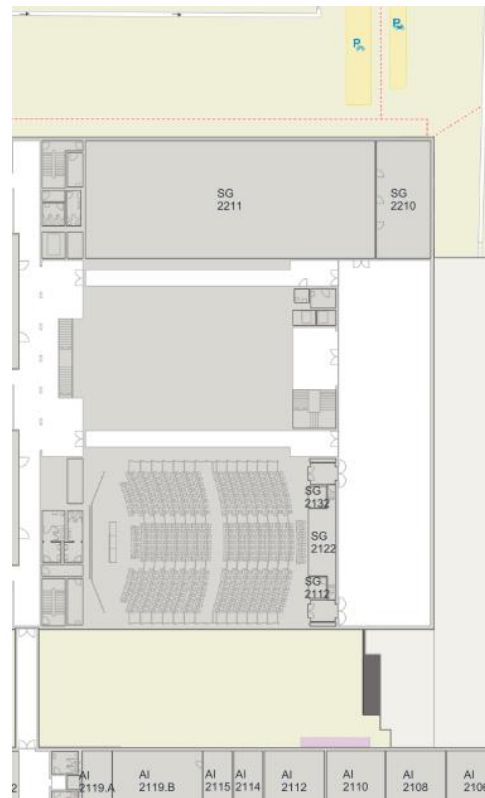
Starling
HotelEPFL/Colladon
bus stop



SG0 (ground floor)



SG1 (first floor)



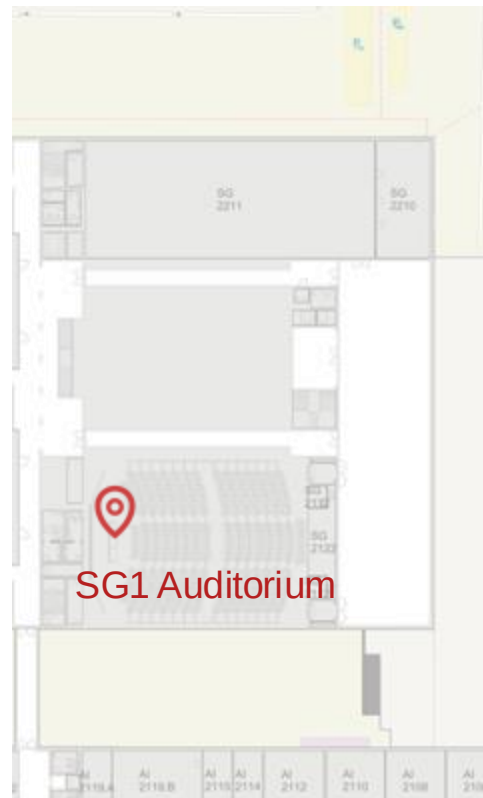
SG2 (second floor)



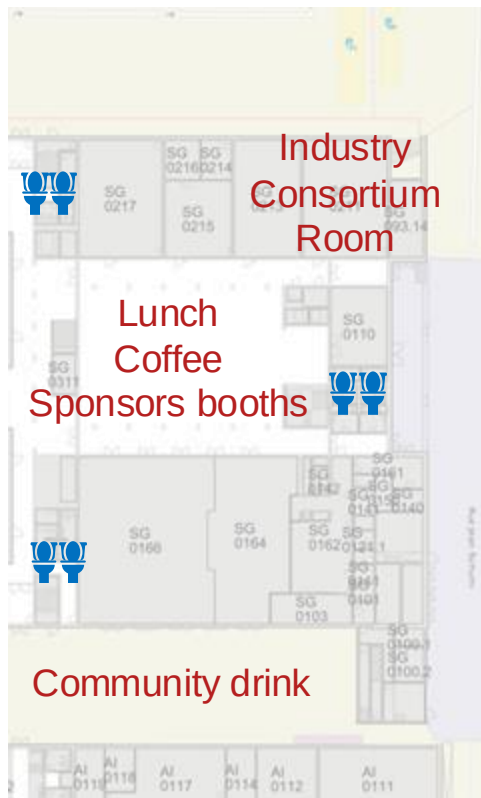
SG0 (ground floor)



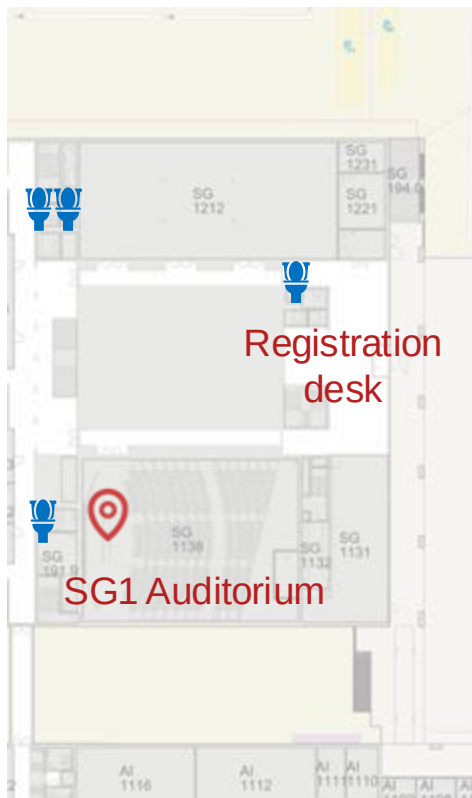
SG1 (first floor)



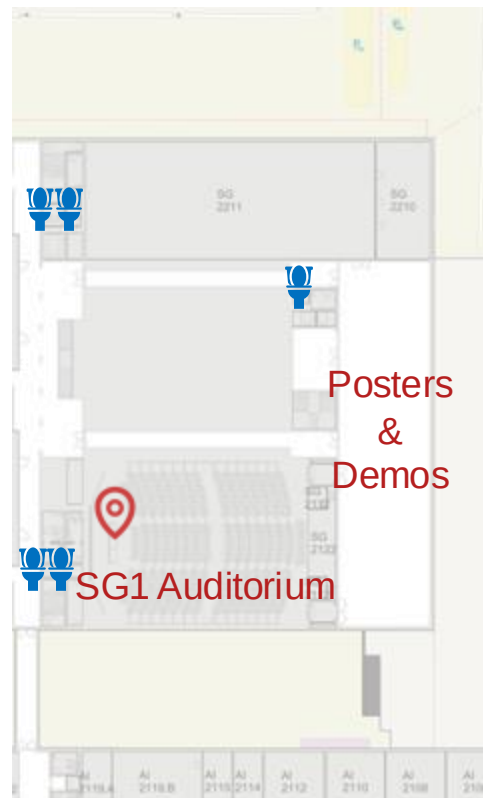
SG2 (second floor)



SG0 (ground floor)



SG1 (first floor)



SG2 (second floor)

- Network: ICCP2024
 - Username: x-iccp2024
 - Password: blapou21 (6 characters + 2 digits)

- Network: Eduroam

Today:

- First keynote
- 7 orals
- 3 invited talks
- First poster and demo session
- Lunch / Women's lunch
- Evening Cocktail Event

Monday, July 22nd		
8.00 am – 9.00 am	 Welcome coffee and Registrations	SG1
9.00 am – 9.30 am	Opening Remarks: Sabine Süsstrunk	
9.30 am – 10.30 am	Keynote A: Davide Scaramuzza: <i>Event Cameras, a New Way of Sensing</i>	
10.30 am – 11.00 am	 Coffee break	SG0
11.00 am – 11.30 am	Invited talk: Amir Zamir: <i>Multimodal Foundation Models</i>	SG1
11.30 am – 12.30 pm Oral session A (11-min talks + 4-min Q&A)	<i>NeMF: Neural Microphysics Fields</i>	
	<i>ThermalNeRF: Thermal Radiance Fields</i>	
	<i>DRAGON: Drone and Ground Gaussian Splatting for 3D Building Reconstruction</i>	
	<i>Z-Splat: Z-Axis Gaussian Splatting for Camera-Sonar Fusion</i>	
12.30 pm – 2.00 pm	 Lunch and Affinity Group Meetings	SG0
2.00 pm – 2.30 pm	Invited talk: Luat T Vuong: <i>Nature-inspired Full-Stokes Polarimetric Camera</i>	SG1
2.30 pm – 3.15 pm Oral session B (11-min talks + 4-min Q&A)	<i>Learned B-Spline Parametrization of Lattice Focal Coding for Monocular RGBD Imaging</i>	
	<i>Task Driven Sensor Layouts - Joint Optimization of Pixel Layout and Network Parameters</i>	
	<i>Stereo-Knowledge Distillation from dpMV to Dual Pixels for Light Field Video Reconstruction</i>	
3.15 pm – 3.45 pm	Invited talk: Emma Alexander: <i>Reverse-Engineering Evolved Cameras</i>	
3.45 pm - 3.55 pm	Poster spotlight session A	
4.00 pm – 4.30 pm	 Coffee break	SG0
4.30 pm – 6.00 pm	Posters & Demos A	SG0+SG2
5.30 pm – 6.30 pm	Outreach	
6.30 pm – 8.00 pm	  Community drink	SG0

Tomorrow:

- Second keynote
- 8 orals
- 3 invited talks
- Second poster and demo session
- Group photo outside of SG1 right before lunch
- Lunch / Industry Consortium Lunch
- Reception at the Olympic Museum

Tuesday, July 23rd		
9.00 am – 10.00 am	Keynote B: Jannick Rolland: <i>Augmented and Virtual Reality: Shaping Tomorrow's World</i>	SG1
10.00 am – 10.30 am	Coffee break	SG0
10.30 am – 11.00 am	Invited talk: Fei Xia: <i>AI-Enhanced Volumetric Optical Microscopy for Faster Imaging in Living Tissues</i>	SG1
	<i>HOLI-1-to-3: Transient-Enhanced Holistic Image-to-3D Generation</i>	
11.00 am – 12.00 pm	Oral session C (11-min talks + 4-min Q&A)	
	<i>Handheld Mapping of Specular Surfaces using Consumer-Grade Flash LiDAR</i>	
	<i>Textureless Deformable Object Tracking with Invisible Markers</i>	
	<i>Deep Phase Coded Image Prior</i>	
12.00 pm – 12.30 pm	Invited talk: Jiamin Wu: <i>Computational light-field imaging for mesoscale intravital microscopy</i>	
12.30 pm – 2.00 pm	Lunch and Industry Consortium Lunch	SG0
2.00 pm – 2.30 pm	Invited Talk: Alexandre Alahi: <i>7 Foundational Principles to Learn Representations</i>	SG1
	<i>Test-time Training for Hyperspectral Image Super-resolution</i>	
2.30 pm – 3.30 pm	Oral session D (11-min talks + 4-min Q&A)	
	<i>Scanning Iridescent Reflectance</i>	
	<i>Efficient Time Sampling Strategy for Transient Absorption Spectroscopy</i>	
	<i>End-to-end Fluorescence Lifetime Imaging with optimized encoding and exposure allocation</i>	
3.30 pm – 3.40 pm	Poster spotlight session B	
3.40 pm – 4.00 pm	Coffee break	SG0
4.00 pm – 5.30 pm	Posters & Demos B	SG0+SG2
6.00 pm – 8.00 pm	Private open visit of the Olympic Museum (7.00 pm – 8.00 pm) Drink on the terrasse of the Olympic Museum	Olympic Museum
8.00 pm – 10.00 pm	Reception at the Olympic Museum restaurant	

Wednesday:

- Third keynote
- 3 orals
- 2 invited talks
- Prizes

Wednesday, July 24th		
9.00 am – 10.00 am	Keynote C: Sylvain Gigan: <i>Imaging with scattered light: wavefront shaping for fluorescence imaging in complex media</i>	SG1
10.00 am – 10.30 am	 Coffee break	SG0
10.30 am – 11.00 am	Invited talk: Ivo Ihrke: <i>Computational Imaging across the Scales</i>	SG1
11.00 am – 11.45 pm Oral session E (11-min talks + 4-min Q&A)	<i>Monte Carlo Path Tracing and Statistical Event Detection for Event Camera Simulation</i>	
	<i>A Practical Framework for Precise Pixel Integration in Simulated Camera Pipelines</i>	
	<i>Deepmarking: Leveraging Adversarial Noise for Membership Inference Attacks</i>	
11.45 am – 12.15 pm	Invited talk: Mark Sheinin: <i>Nothing Lasts Forever: Harnessing Ephemeral Signals for Dynamic Scene Perception</i>	
12.15 pm – 12.45 pm	Closing Remarks, Best paper award, Poster and Demo prize: Sabine Süsstrunk	
12.45 pm – 2.00 pm	 Lunch / Committee lunch / End of the conference	SG0

ICCP2024 Keynote Speakers



Davide Scaramuzza
University of Zurich
*Event Cameras, a New Way
of Sensing*
Today at 9.30am



Jannick Rolland
Rochester University
*Augmented and Virtual
Reality: Shaping Tomorrow's
World*
Tomorrow at 9.00am

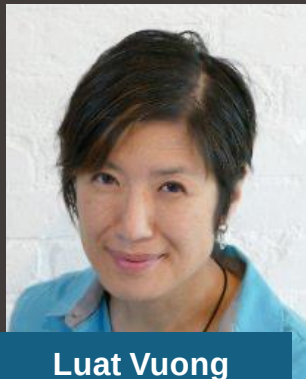


Sylvain Gigan
ENS – CNRS
*Imaging with scattered light:
wavefront shaping for
fluorescence imaging in
complex media*
Wednesday at 9.00am

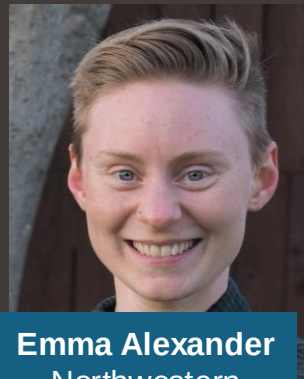
ICCP2024 Invited Speakers



Amir Zamir
EPFL



Luat Vuong
UC Riverside



Emma Alexander
Northwestern
University



Ivo Ihrke
Siegen University



Fei Xia
ENS Paris



Jiamin Wu
Tsinghua
University



Alexandre Alahi
EPFL



Mark Sheinin
Weizmann Institute
of Science

- Spotlight poster/demo videos are already uploaded, recorded orals after the conference.

YouTube CH

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ICCP

@iccp-conference · 413 subscribers · 97 videos

Official channel of the IEEE International Conference on Computational Photography. ...more

iccp-conference.org and 2 more links

Customize channel Manage videos

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Latest Popular Oldest

Tiltable phase modulator using binary tilts

2:00

Optimal DOE for 2D target

2:00

Poster 7. Exponentially-wide etendue displays using a tilting cascade

91 views · 1 year ago

Paper 6. Analyzing phase masks for wide etendue holographic displays

129 views · 1 year ago

Poster 1. D-Flat: A Differentiable Flat-Optics Framework for...

765 views · 1 year ago

Poster 20. The Implicit Values of a Good Hand Shake: Handheld Multi-Frame Neural Depth Refinement

311 views · 1 year ago

Poster 44. Pupil-aware Holography

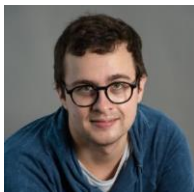
252 views · 1 year ago

Poster 2. Single-Photon Structured Light

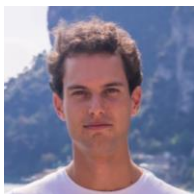
292 views · 1 year ago

<https://www.youtube.com/@iccp-conference>

- X (Twitter)
- YouTube Community
- Facebook



Social Media chair
Martin N Everaert

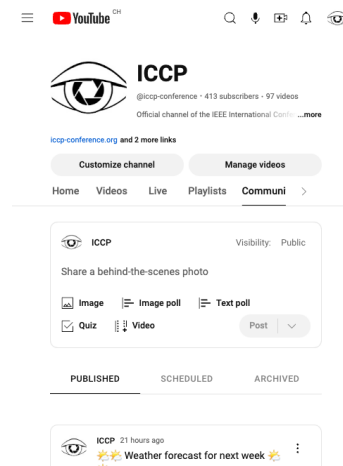
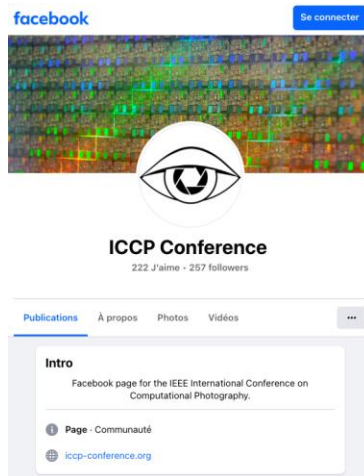


Photographer (volunteer)
Federico Stella



https://x.com/ICCP_conference

<https://www.facebook.com/ieeEICCP/>



<https://www.youtube.com/@iccp-conference>

ICCP Posters/Demos

2 poster & demo sessions

- Today: 4:30-6:00pm (+outreach: 5.30-6.30pm)
- Tomorrow: 4:00-5:30pm

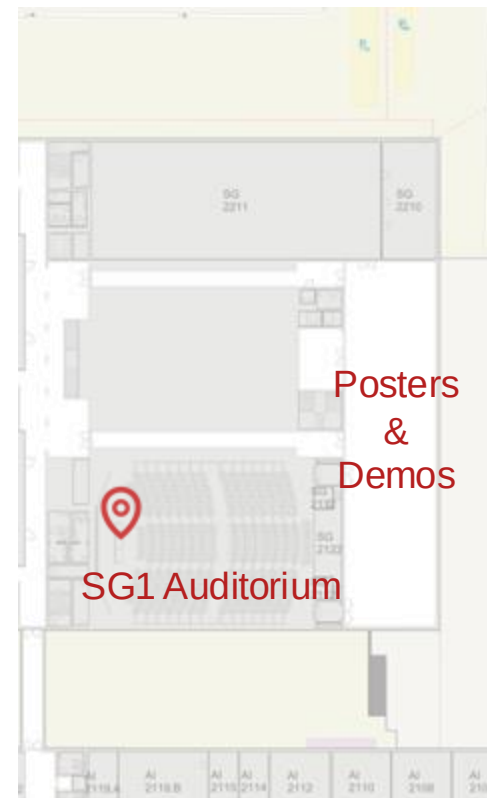
Set up poster during the prior coffee break

Uninstall before the next poster session



Poster & Demo chairs

Kristina Monakhova, Adithya Pediredla, Eric Bezzam



SG2 (second floor)

ICCP Poster/Demo Awards

2 poster sessions

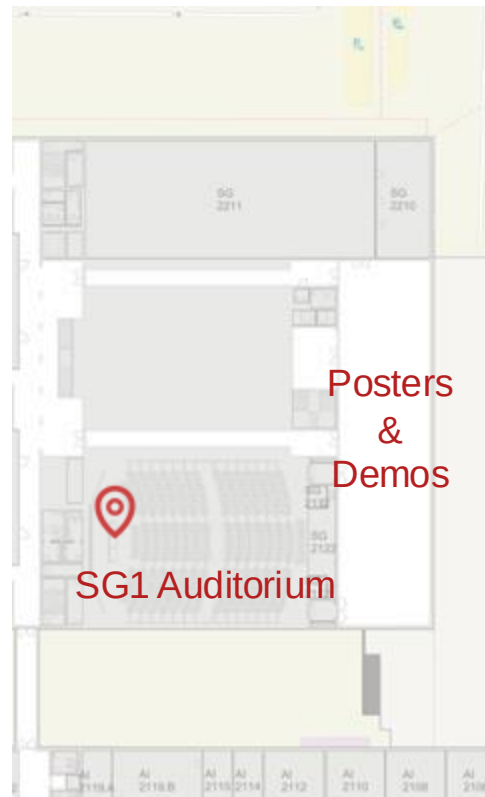
- Today: 4:30-6:00pm (+outreach: 5.30-6.30pm)
- Tomorrow: 4:00-5:30pm

Best poster/demo award (from a list of shortlisted posters)

Stay tuned for details on voting



Poster & Demo chairs
Kristina Monakhova, Adithya Pediredla, Eric Bezzam

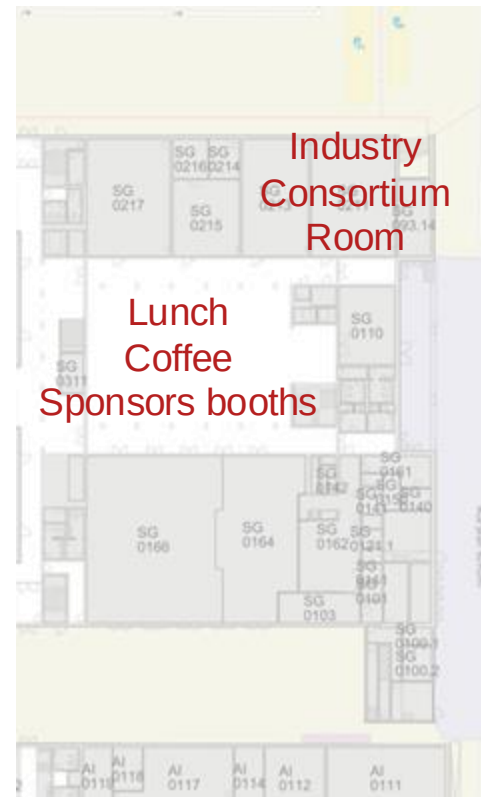


SG2 (second floor)

- Goal: Foster stronger connections between academia and industry
- Targeted networking event to help students and postdocs prepare for careers in industry
- Tuesday during lunch time



Industry consortium chair
Dengxin Dai

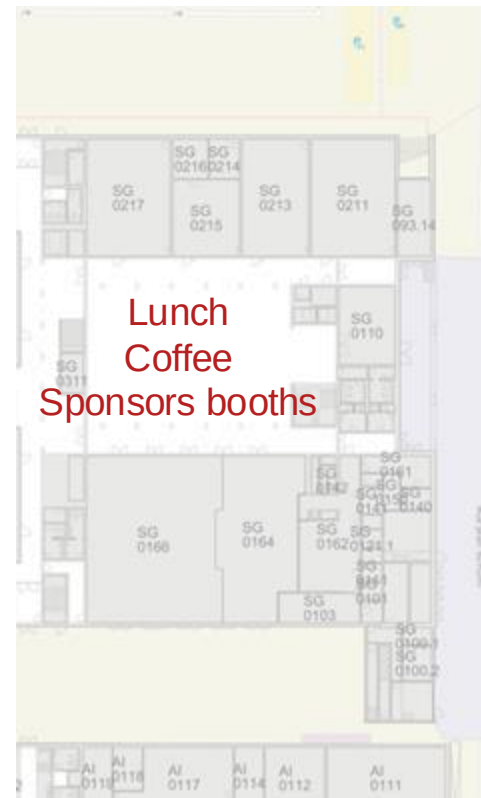


SG0 (ground floor)

- Your lunch is served in recyclable containers, please bring them back to the lunch desk.
- Water bottles are provided in your welcoming kit, please use those instead of cups.
 - There are 2 water dispenser.
 - Stickers and/or water-proof pens available at the reception.



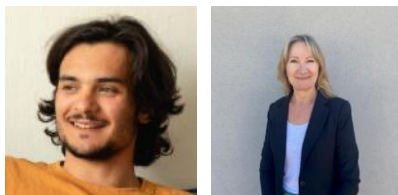
Local arrangements chairs
Raphaël Achddou, Nicoletta Isaac



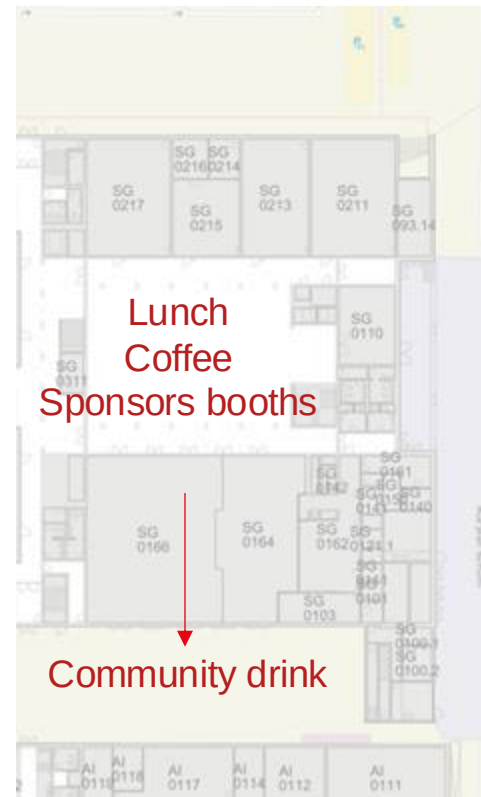
SG0 (ground floor)

Tonight 6:30-8:00pm

- Evening Cocktail Event



Local arrangements chairs
Raphaël Achddou, Nicoletta Isaac



SG0 (ground floor)

ICCP2024 Reception at the Olympic Museum

Tomorrow 6:00-10:00pm

- Private Visit of the museum (starting at 6 pm)
- Drink on the Olympic Museum Terrasse (starting at 7pm)
- Dinner in the Olympic Museum restaurant, top floor (starting at 8 pm)



Local arrangements chairs
Raphaël Achddou, Nicoletta Isaac



© <https://www.lausanne-tourisme.ch/en/explore/the-olympic-museum/>

How to go to the Olympic Museum?

5:24 PM	EPFL
	 m1 Lausanne-Flon, pl. de l'Europe ✓ 13 min (9 stops)
5:37 PM 5:42 PM	Lausanne-Flon
	 m2 Lausanne, Ouchy-Olympique ✓ 8 min (5 stops)
5:50 PM	Ouchy-Olympique
	 Walk ✓ About 12 min, 800 m
6:02 PM	The Olympic Museum Quai d'Ouchy 1, 1006 Lausanne

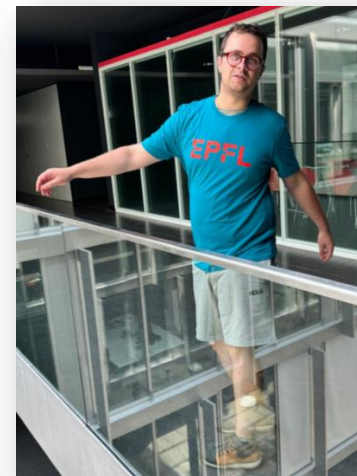
5:33 PM	Ecublens VD, EPFL/Colladon
	 T1 Lausanne, Blécherette ✓ 18 min (13 stops)
5:51 PM	Cour
	 Walk ✓ About 13 min, 950 m
6:04 PM	Quai d'Ouchy 1 1006 Lausanne

5:32 PM	EPFL
	 m1 Lausanne-Flon, pl. de l'Europe ✓ 5 min (4 stops)
5:37 PM 5:40 PM	Bourdonnette
	 24 Lausanne, Tour Haldimand-Lac ✓ 17 min (10 stops)
5:57 PM	Parc Musée Olympique
	 Walk ✓ About 6 min, 300 m
6:03 PM	Quai d'Ouchy 1 1006 Lausanne

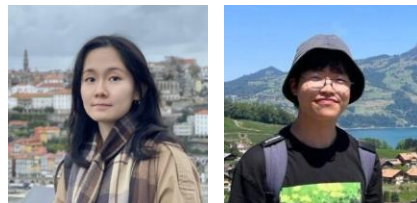
5:32 PM	EPFL
	 m1 Lausanne-Flon, pl. de l'Europe ✓ 11 min (8 stops)
5:43 PM	Vigie
	 Walk ✓ About 2 min
5:49 PM	Cécil
	 21 Paudex, Verrière ✓ 12 min (6 stops)
6:01 PM	Musée Olympique
	 Walk ✓ About 2 min, 180 m
6:03 PM	Quai d'Ouchy 1 1006 Lausanne

Many public transportation options (30-40 minutes)

Just follow any volunteer 😊



- Create together a photo gallery that captures the experiences you have during the conference
- Your goal is to WIN 😊
- Upload your photos (max. 3) by Tuesday evening
- Vote on Wednesday morning



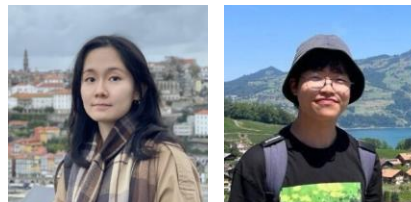
Website chairs
Dongqing Wang, Yufan Ren

Instructions on ICCP website:



PHOTOGRAPHY COMPETITION

ICCP2024 Photography competition



Website chairs
Dongqing Wang, Yufan Ren

- If you are interested in hosting ICCP 2026, please talk to Ashok, Wolfgang, or Sabine during the conference and/or send us a mail:
 - vashok@rice.edu
 - wolfgang.heidrich@kaust.edu.sa
 - sabine.susstrunk@epfl.ch

ICCP2024 Technical Program

48 submissions, 18 accepted

- 12 accepted to ICCP proceedings (ICCP orals)
- 6 accepted to TPAMI (also ICCP orals)



Program chairs
Ashok Veeraraghavan, Roarke Horstmeyer,
Wolfgang Heidrich

IEEE TRANSACTIONS ON

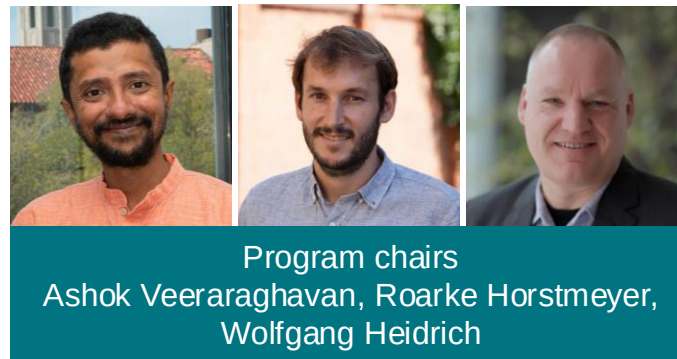
PATTERN ANALYSIS AND
MACHINE INTELLIGENCE



PAMI chair
Boxin Shi

ICCP2024 Review Process

- 3-4 initial reviews / paper
- 1 week rebuttal period
- Consensus decisions of reviewers on conditional accept vs. reject
- TPAMI decisions by chairs
- Shepherding for final acceptance by selected PC members

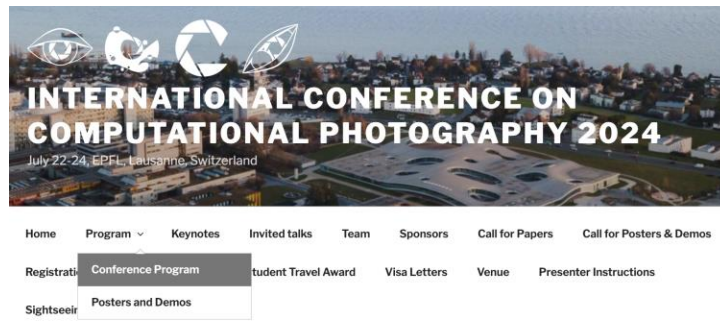


- Chairs did not submit papers

- Download requires password (emailed to all registered attendees)
- Can be downloaded up to Wednesday evening only
- After: IEEE Computer Society Digital Library and IEEE Xplore



Publication chairs
Vivek Boominathan, Akshat Dave



CONFERENCE PROGRAM

- Password:
 - `proc_iccp24`



- The awards will be given at the closing ceremony on Wednesday at noon:
 - Best paper
 - Best poster
 - Best demo

- And the photographic competition award!

Welcome To ICCP2024



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July 22-24, 2024